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[INSPIRE HEP](#) | [arXiv](#) | [Google Scholar](#)

Preprints and Submitted Manuscripts

- [1] J. Komijani, “Noise scheduling and linear dynamics in diffusion models on Lie groups,” [\[arXiv:2605.17326\]](#).
[INSPIRE-HEP entry](#)
- [2] J. Komijani, M. Marinkovic and L. Turgut, “Diffusion model for SU(N) gauge theories,” [\[arXiv:2605.06134\]](#).
[INSPIRE-HEP entry](#)
- [3] O. Vega, J. Komijani, A. El-Khadra and M. Marinkovic, “Group-Equivariant Diffusion Models for Lattice Field Theory,” [\[arXiv:2510.26081\]](#).
[INSPIRE-HEP entry](#)

Peer-reviewed Publications

- [1] A. Manzi *et al.*, “interTwin: Advancing Scientific Digital Twins through AI, Federated Computing and Data”, *Future Generation Computer Systems*, **179**, 108312 (2026).
- [2] A. Altherr *et al.*, “Comparing QCD+QED via full simulation versus the RM123 method: U-spin window contribution to a_μ^{HVP} ,” *JHEP* **10**, 158 (2025) [[arXiv:2506.19770](#)].
[INSPIRE-HEP entry](#)
- [3] R. Aliberti *et al.*, “The anomalous magnetic moment of the muon in the Standard Model: an update,” *Phys. Rept.* **1143**, 1-158 (2025) [[arXiv:2505.21476](#)].
[INSPIRE-HEP entry](#)
- [4] C. M. Bender and J. Komijani, “Addendum: Painlevé transcendents and PT-symmetric Hamiltonians,” *J. Phys. A: Math. Theor.* **55**, 109401 (2022) [[arXiv:2107.04935](#)].
[INSPIRE-HEP entry](#)
- [5] J. Komijani, “First-order nonlinear eigenvalue problems involving functions of a general oscillatory behavior,” *J. Phys. A: Math. and Theor.* **54**, 465202 (2021) [[arXiv:2107.02475](#)].
[INSPIRE-HEP entry](#)
- [6] J. Komijani, P. Petreczky and J. H. Weber, “Strong coupling constant and quark masses from lattice QCD,” *Prog. Part. Nucl. Phys.* **113**, 103788 (2020) [[arXiv:2003.11703](#)].
[INSPIRE-HEP entry](#)
- [7] L. J. Cooper *et al.*, “ $B_c \rightarrow B_{s(d)}$ form factors from lattice QCD,” *Phys. Rev. D* **102**, 014513 (2020) [[arXiv:2003.00914](#)].
[INSPIRE-HEP entry](#)
- [8] C. M. Bender, J. Komijani, Q. Wang, “Nonlinear eigenvalue problems for generalized Painlevé equations,” *J. Phys. A: Math. and Theor.* **52**, 315202 (2019) [[arXiv:1903.10640](#)].
[INSPIRE-HEP entry](#)

- [9] C. T. H. Davies *et al.*, “Determination of the quark condensate from heavy-light current-current correlators in full lattice QCD,” *Phys. Rev. D* **100**, 034506 (2019) [arXiv:1811.04305].
INSPIRE-HEP entry
- [10] N. Brambilla, H. S. Chung and J. Komijani, “Inclusive decays of η_c and η_b at NNLO with large n_f resummation,” *Phys. Rev. D* **98**, 114020 (2018) [arXiv:1810.02586].
INSPIRE-HEP entry
- [11] A. Bazavov *et al.*, “ $|V_{us}|$ from $K_{\ell 3}$ decay and four-flavor lattice QCD,” *Phys. Rev. D* **99**, 114509 (2019) [arXiv:1809.02827].
INSPIRE-HEP entry
- [12] A. Bazavov *et al.*, “Up-, down-, strange-, charm-, and bottom-quark masses from four-flavor lattice QCD,” *Phys. Rev. D* **98**, 054517 (2018) [arXiv:1802.04248].
INSPIRE-HEP entry
- [13] A. Bazavov *et al.*, “ B - and D -meson leptonic decay constants from four-flavor lattice QCD,” *Phys. Rev. D* **98**, 074512 (2018) [arXiv:1712.09262].
INSPIRE-HEP entry
- [14] N. Brambilla, J. Komijani, A.S. Kronfeld, A. Vairo, “Relations between Heavy-light Meson and Quark Masses,” *Phys. Rev. D* **97**, 034503 (2018) [arXiv:1712.04983].
INSPIRE-HEP entry
- [15] J. Komijani, “A discussion on leading renormalon in the pole mass,” *JHEP* **1708**, 062 (2017) [arXiv:1701.00347].
INSPIRE-HEP entry
- [16] J.A. Bailey *et al.*, “ $B \rightarrow Kl^+l^-$ decay form factors from three-flavor lattice QCD,” *Phys. Rev. D* **93**, 025026 (2016) [arXiv:1509.06235].
INSPIRE-HEP entry
- [17] J.A. Bailey *et al.*, “ $|V_{ub}|$ from $B \rightarrow \pi \ell \nu$ decays and (2+1)-flavor lattice QCD,” *Phys. Rev. D* **92**, 014024 (2015) [arXiv:1503.07839].
INSPIRE-HEP entry
- [18] J.A. Bailey *et al.*, “ $B \rightarrow D \ell \nu$ form factors at nonzero recoil and $|V_{cb}|$ from 2+1-flavor lattice QCD,” *Phys. Rev. D* **92**, 034506 (2015) [arXiv:1503.07237].
INSPIRE-HEP entry
- [19] A. Bazavov *et al.*, “Gradient flow and scale setting on MILC HISQ ensembles,” *Phys. Rev. D* **93**, 094510 (2016) [arXiv:1503.02769].
INSPIRE-HEP entry
- [20] C. M. Bender and J. Komijani, “Painlevé Transcendents and $\mathcal{PT}$ -Symmetric Hamiltonians,” *J. Phys. A: Math. and Theor.* **48**, 475202 (2015) [arXiv:1502.04089].
INSPIRE-HEP entry
- [21] A. Bazavov *et al.*, “Charmed and light pseudoscalar meson decay constants from four-flavor lattice QCD with physical light quarks,” *Phys. Rev. D* **90**, 074509 (2014) [arXiv:1407.3772].
INSPIRE-HEP entry

- [22] C. M. Bender, A. Fring and J. Komijani, “Nonlinear Eigenvalue Problems,” *J. Phys. A: Math. and Theor.* **47**, 235204 (2014) [arXiv:1401.6161].
INSPIRE-HEP entry
- [23] C. Bernard and J. Komijani, “Chiral Perturbation Theory for All-Staggered Heavy-Light Mesons,” *Phys. Rev. D* **88**, 094017 (2013) [arXiv:1309.4533].
INSPIRE-HEP entry
- [24] A. Bazavov *et al.*, “Lattice QCD ensembles with four flavors of highly improved staggered quarks,” *Phys. Rev. D* **87**, 054505 (2013) [arXiv:1212.4768].
INSPIRE-HEP entry
- [25] J. Komijani, J. Rashed-Mohassel and A. Mirkamali, “Dyadic Green Functions for a Dielectric Layer on a PEMC Plane,” *Progress in Electromagnetics Research M* **6**, pp. 9-22 (2009).
- [26] J. Komijani and J. Rashed-Mohassel, “Symmetrical Properties of Dyadic Green’s Functions for Mixed Boundary Conditions and Integral Representations of the Electric Fields for Problems Involving a PEMC,” *IEEE Transactions on Antennas and Propagation*, vol. 57, issue 10, pp. 3199-3204 (2009).

Selected Conference Proceedings

- [1] J. Komijani and M. K. Marinkovic, “Normalizing flows for SU(N) gauge theories employing singular value decomposition,” *PoS Lattice 2024*, 050 [arXiv:2501.18288].
INSPIRE-HEP entry
- [2] J. Komijani and M. K. Marinkovic, “Generative models for scalar field theories: how to deal with poor scaling?” *PoS Lattice 2022*, 019 [arXiv:2301.01504].
INSPIRE-HEP entry
- [3] A. Altherr *et al.*, “Strange and charm contributions to the HVP from C* boundary conditions,” *PoS Lattice 2022*, 281 [arXiv:2301.04385].
INSPIRE-HEP entry
- [4] A. Altherr *et al.*, “Hadronic vacuum polarization with C* boundary conditions,” *PoS Lattice 2022*, 312 [arXiv:2212.11551].
INSPIRE-HEP entry
- [5] G. Abbiendi *et al.*, “Space-like and Time-like determination of the Hadronic Leading Order contribution to the Muon $g - 2$,” *Mini-Proceedings of the STRONG2020 Virtual Workshop* [arXiv:2201.12102].
INSPIRE-HEP entry
- [6] L. Cooper *et al.*, “ $B_c \rightarrow B_{s(d)}$ form factors,” *PoS Lattice 2019*, 103 [arXiv:1911.04282].
INSPIRE-HEP entry
- [7] R. Li *et al.*, “ D meson Semileptonic Decay Form Factors at $q^2 = 0$,” *PoS Lattice 2018*, 269 [arXiv:1901.08989].
INSPIRE-HEP entry
- [8] A. Bazavov *et al.*, “ B - and D -meson leptonic decay constants and quark masses from four-flavor lattice QCD” [arXiv:1810.00250].
INSPIRE-HEP entry

- [9] C. M. Bender, J. Komijani, Q. Wang, “Nonlinear eigenvalue problems,” In: *Resurgence, Physics and Numbers, CRM Series, vol 20, pp. 67-89, 2017.*
- [10] T. Primer *et al.*, “ D meson semileptonic form factors with HISQ valence and sea quarks,” *PoS Lattice 2016*, 305.
[INSPIRE-HEP entry](#)
- [11] J. Komijani *et al.*, “Decay constants f_B and f_{B_s} and quark masses m_b and m_c from HISQ simulations,” *PoS Lattice 2016*, 294 [arXiv:1611.07411].
[INSPIRE-HEP entry](#)
- [12] E. Gamiz *et al.*, “Kaon semileptonic decays with $N_f=2+1+1$ HISQ fermions and physical light-quark masses,” *PoS Lattice 2016*, 286 [arXiv:1611.04118].
[INSPIRE-HEP entry](#)
- [13] T. Primer *et al.*, “ D -Meson Semileptonic Form Factors at Zero Momentum Transfer in $(2+1+1)$ -Flavor Lattice QCD,” *PoS Lattice 2015*, 338 [arXiv:1511.04000].
[INSPIRE-HEP entry](#)
- [14] A. Bazavov *et al.*, “Decay constants f_B and f_{B_s} from HISQ simulations,” *PoS Lattice 2015*, 331 [arXiv:1511.02294].
[INSPIRE-HEP entry](#)
- [15] S. Basak *et al.*, “Electromagnetic effects on the light hadron spectrum,” *J. Phys. Conf. Ser.* 640, 012052 (2015) [arXiv:1510.04997].
[INSPIRE-HEP entry](#)
- [16] A. Bazavov *et al.*, “Charmed and light pseudoscalar meson decay constants from HISQ simulations,” *PoS Lattice 2014*, 382 [arXiv:1411.2667].
[INSPIRE-HEP entry](#)
- [17] A. Bazavov *et al.*, “Gradient Flow Analysis on MILC HISQ Ensembles,” *PoS Lattice 2014*, 090 [arXiv:1411.0068].
[INSPIRE-HEP entry](#)
- [18] S. Basak *et al.*, “Finite-volume effects and the electromagnetic contributions to kaon and pion masses,” *PoS Lattice 2014*, 116 [arXiv:1409.7139].
[INSPIRE-HEP entry](#)
- [19] A. Bazavov *et al.*, “The D_s , D^+ , B_s , and B^+ decay constants from 2+1 flavor lattice QCD,” *PoS Lattice 2013*, 394 [arXiv:1403.6796].
[INSPIRE-HEP entry](#)
- [20] A. Bazavov *et al.*, “Charmed and strange pseudoscalar meson decay constants from HISQ simulations,” *PoS Lattice 2013*, 405 [arXiv:1312.0149].
[INSPIRE-HEP entry](#)
- [21] J. Komijani and C. Bernard, “Staggered chiral perturbation theory for all-staggered heavy-light mesons,” *PoS Lattice 2012*, 199 [arXiv:1211.0785].
[INSPIRE-HEP entry](#)

- [22] J. Komijani, A. Mirkamali and J. Nateghi, “Combining multiple knife-edge diffraction and ground reflections for terrain path loss diffraction,” *Proceedings of the Fourth European Conference on Antennas and Propagation, 2010*.
[IEEE](#)
- [23] J. Komijani, J. Rashed-Mohassel and H. Hadidian-Moghadam, “Green function for a horizontal source over a dielectric slab with a PEMC ground,” *Proceedings of the Third European Conference on Antennas and Propagation, 2009*.
[IEEE](#)